



### Dolphins have weaponised vaginas

Bottlenose dolphins have evolved complicated, folded vaginas that make it difficult for unwanted males to fertilise eggs <http://digit.in/DolphWpn>



### Real life quantum communication

A quantum-encrypted message was beamed through the air between two buildings for the first time <http://digit.in/QntmComm>

# W

here do you go when you sleep? Is it still even you in there when you're sleeping? When you wake

up, is it the same you that went to sleep a few hours ago, or have you changed a little? While we're making headway in understanding the conscious mind through neurobiology, psychology and psychiatry, we're still not sure about the unconscious mind. No, wait, let's rephrase that... because the term "unconscious mind" is something philosophers and psychologists have used to describe what they believe to be the lowest functioning layer of the brain, on which the higher states of consciousness are built. Instead, let's call it the *minimally-conscious* mind, because this isn't an article about psychology...

### MINIMALLY-CONSCIOUS?

We just made up the term, to be honest, and then after checking it, we found that it too is already used elsewhere in medicine, as most word combinations already are. Thus, we should clarify that what we mean by this is very specifically the non-waking human mind. Whether in a deep slumber, under general anaesthesia, in a coma (medically induced or otherwise), and various vegetative states, including people we consider brain dead (or legally dead). In essence, it's a non-scientific, layman's definition that we're just going to run with for this article, so hold off on those angry letters, if you happen to be a psychology, psychiatry or medical practitioner.

While we're inventing our own word salad to describe the scope of the article, that's where our trickery ends. Everything from here on out is strictly going to be according to biology and medicine, and to start off with, let's look at the most common state we all partake in...

### SLEEP

A normal person spends about a third of a lifetime doing this. Thus, if you live to be 75, you will have spent 25

years of those fast asleep. The type of sleep we humans experience is thought to be subdivisible into two – REM (rapid eye movement) and non-REM sleep. A normal human cycles through REM and the three different types of non-REM sleep through the night. The three different kinds of non-REM sleep are basically called the three stages of non-REM sleep.

Stage 1 is the process of falling asleep, and the change in the brain from waking to sleeping. Stage 2 involves even further slowing of heart-beat and breathing, and brain activity slows down considerably. Stage 3 non-REM sleep is the deep sleep we experience that is very important. Stage 3 non-REM sleep is why we can feel refreshed after a "good sleep", and it's the period that's the hardest to wake you from.

REM sleep is the period when most of the dreams we have (which we can remember) occur. It's called REM because although our eyes are closed (usually) during this mode of sleep, our eyes move back and forth rapidly, and brain wave function is very similar to our waking selves. We could probably write a book on just sleep alone, but it's not the scope of this article. What matters to the context of this article, is that the brain itself goes through a phase of "minimal consciousness" (or deep sleep) and yet there is still always brain activity all through our sleep. In short, we don't know where we (as in, our conscious minds) go when we sleep, but we hope to be able to find out in the near future.

### COMA

Jumping straight into the real reason for this article... we've always been intrigued by comas. Heck, the movies especially would have you believe that comas are just deep sleeps that people can wake up from after years, or even decades, as if nothing happened. Of course most movies depict comas totally unrealistically, and you don't want to get us started on movies not doing their research properly!

So what distinguishes a coma from all other minimally-conscious states? For starters, comas are a state of unconsciousness where the normal wake-sleep cycle is totally disrupted. It's simply a state where the person cannot be awoken normally, and doesn't respond to light, sound, pain, etc., and shows no response to any stimuli whatsoever.

A medical patient can also be in a coma that was initiated by doctors on purpose. Known as a medically induced coma, this is effectively a form of general anaesthesia that's administered to patients to put them in a comatose state. This is often done when it is beneficial for the patient to be totally immobilised and unconscious to aid in the healing process, or when injuries are so severe that being awake would be excruciatingly painful for the patient.

But how significant does the impairment of your faculties have to be, before you will be considered to be in a coma? And is there a way to tell if one patient is in a deeper coma than another? The Glasgow Coma Scale (GCS) is exactly that ranking system, and it was published by Bryan Jennett and Graham Teasdale in 1974, when they were professors of Neurosurgery at the University of Glasgow's Southern General Hospital. They divided all of humanity into a scale ranging from 3 to 14 (later modified up to 15). The modern GCS is presented in the table on the next page.

As you can see, eye movement is scored from 1 to 4, verbal ability from 1 to 5, and motor movement from 1 to 6. A healthy individual would have a GCS of 15, and someone in a severe coma or even a dead person would have a score of 3.

Everything about the eyes and speech is pretty straightforward, and there's nothing that really needs any explaining. When testing all three of the parameters, the first test done is not the lowest level tests, but the highest. Thus, when observing a patient, if they open their eyes spontaneously, they automatically score 4 for eyes.